

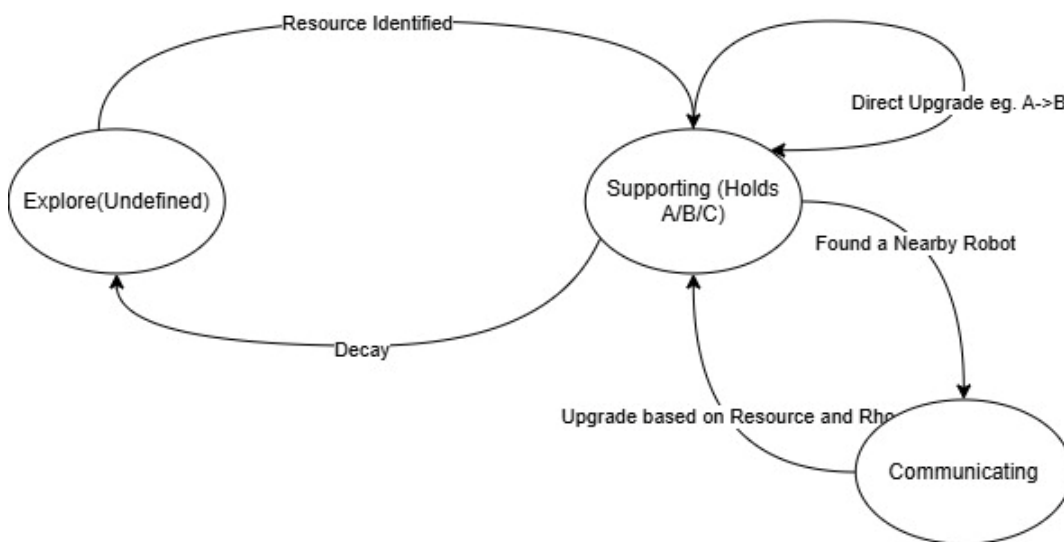
Ant Food Source Selection

Design Decisions:

Microscopic Model Design:

Robot States:

The below image shows various states of the Microscopic Robot model. Explanation of each state is given below.



States of Microscopic output:

$S = \{U, A, B, C\}$

Where,

- U = Undecided
- A = Support Resource A
- B = Support Resource B
- C = Support Resource C

Robots initially hold the value U. They may transition from $U \rightarrow A/B/C$ after resource discovery and evaluation.

Through local communication, robots may switch between resource-support states. Over time, a robot may abandon its current preference and transition from $A/B/C \rightarrow U$.

Given Resource sites $R = \{A, B, C\}$

And Quality of each resource be between, $R_i \in [0,1]$

Each Robot decides the resource it supports based on the resource it encounters as below.

$$Si^* = \operatorname{argmax} Ri$$

This value will get updated whenever a new resource is encountered

Local Communication:

Robots shall communicate with its neighbours when they are within a certain distance C_d between its peers.

Based on the communication the robot may update its opinion to support a resource. Below are few scenarios how the communication influence resource support.

Let Resource Value = $\{U: 0, A:0.7, B:0.3\}$

When $d_{ij} \leq C_d$

Case A:

Robot 1 – Supports A: then $S1v = 0.7$

Robot 2 – Supports B: then $S2v = 0.3$

When they communicate the probability of opinion update is determined by the social influence parameter and $\max(\text{Resource Values})$

$$P(\text{Robot 1}(A \rightarrow B)) = \begin{cases} \rho, & \text{if } S2v > S1v \\ 0, & \text{if } S2v \leq S1v \end{cases}$$

$$P(\text{Robot 2}(B \rightarrow A)) = \begin{cases} \rho, & \text{if } S1v > S2v \\ 0, & \text{if } S1v \leq S2v \end{cases}$$

Case B:

Robot 1 – Supports A: then $S1v = 0.7$

Robot 2 – Undefined U: then $S2v = 0$

$$P(\text{Robot 2}(U \rightarrow A)) = \begin{cases} \rho, & \text{if } S1v > S2v \\ 0, & \text{if } S1v \leq S2v \end{cases}$$

Where,

$$\rho = 0.2$$

$S1v$ = Value of resource Robot 1 supports

$S2v$ = Value of resource Robot 2 supports

Abandonment:

In order to avoid robots from becoming permanently locked into an early decision, the Abandonment step is taken based on a Decay / loss of confidence.

Undefined Robots doesn't undergo Abandonment.

$$P_{\text{abandon}}(S_i \rightarrow U) = \gamma \Delta t$$

Where,

$$\gamma = 0.02$$

$$\Delta t = 1s$$